

14

Anti-differentiation and introduction to integral calculus

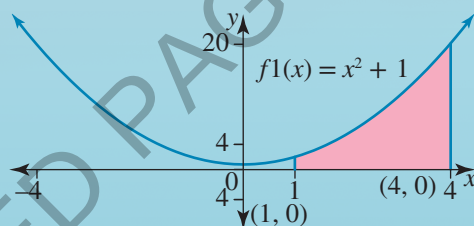
- 14.1 Kick off with CAS
- 14.2 Anti-derivatives
- 14.3 Anti-derivative functions and graphs
- 14.4 Applications of anti-differentiation
- 14.5 The definite integral
- 14.6 Review **eBookplus**



14.1 Kick off with CAS

Integral calculus

- 1 Using CAS technology, find the derivative of $y = x^2$ by using $\frac{d(x^2)}{dx}$.
- 2 Using CAS technology, find the template for the integral and calculate $\int 2x dx$.
- 3 Comparing the results in questions 1 and 2, what do you notice?
- 4 Using CAS technology, find the derivative of $y = 5x^3 + 2x$ by using $\frac{d(5x^3 + 2x)}{dx}$.
- 5 Using CAS technology, calculate $\int (15x^2 + 2) dx$.
- 6 Comparing the results in questions 4 and 5, what do you notice?
- 7 Using CAS technology, sketch the graph of $f(x) = x^2 + 1$.
- 8 Draw vertical lines from the x -axis to the graph at $x = 1$ and at $x = 4$. Estimate the area enclosed by the x -axis, the vertical lines and the graph.



- 9 Using CAS technology, calculate $\int_1^4 (x^2 + 1) dx$.
- 10 What do you notice about the answers to questions 8 and 9?



14.2 Anti-derivatives

study on

Units 1 & 2

AOS 3

Topic 3

Concept 1

Anti-derivatives

Concept summary

Practice questions

Calculus is made up of two parts: differential calculus and integral calculus.

Differential calculus arose from the need to measure an instantaneous rate of change, or gradient; integral calculus arose from the need to measure the area enclosed between curves. The fundamental theorem of calculus which established the connection between the two branches of calculus is one of the most important theorems in mathematics. Before we can gain some understanding of this theorem, we must first consider the reverse operation to differentiation.

Anti-derivatives of polynomial functions

Anti-differentiation is the reverse process (or 'undoing' process) to differentiation.

The derivative of x^2 is $2x$; hence, reversing the process, an anti-derivative of $2x$ is x^2 .

The expression 'an' anti-derivative is of significance in the above example. For any

constant c , $\frac{d}{dx}(x^2 + c) = 2x$, so 'the' anti-derivative of $2x$ is $x^2 + c$. The constant c is referred to as an arbitrary constant.

If $\frac{dy}{dx} = 2x$, then $y = x^2 + c$. In this family of parabolas $y = x^2 + c$, each member has identical shape and differs only by the amount of vertical translation each has undergone. Each curve has the same gradient function of $\frac{dy}{dx} = 2x$. Some members of the family of anti-derivatives are shown in the diagram.

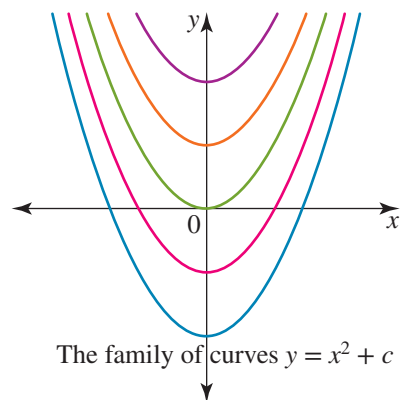
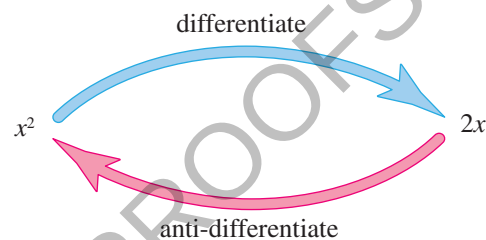
Given $\frac{dy}{dx}$ or $f'(x)$, the process of obtaining y or $f(x)$ is called anti-differentiation. The process is also referred to as finding the **primitive function**, given the gradient function.

The basic rule for anti-differentiation of polynomials

By studying the patterns in the following examples, the basic rule of anti-differentiation can be deduced. The values in the table can be verified by differentiating the polynomial y .

The rule the table illustrates is:

If $\frac{dy}{dx} = x^n, n \in \mathbb{N}$
then $y = \frac{1}{n+1}x^{n+1} + c$, where c is an arbitrary constant.



The family of curves $y = x^2 + c$

$\frac{dy}{dx}$	y
5	$5x + c$
$2x$	$x^2 + c$
$3x^2$	$x^3 + c$
x^2	$\frac{1}{3}x^3 + c$
x^3	$\frac{1}{4}x^4 + c$

This rule can be verified by differentiating y with respect to x .

The linearity properties allow anti-derivatives of sums and differences of polynomial terms to be calculated. For example, if $f'(x) = a_1 + a_2x + a_3x^2 + \dots + a_nx^{n-1}$, $n \in N$, then

$$\begin{aligned} f(x) &= a_1x + a_2\frac{x^2}{2} + a_3\frac{x^3}{3} + \dots + a_n\frac{x^n}{n} + c \\ &= a_1x + \frac{1}{2}a_2x^2 + \frac{1}{3}a_3x^3 + \dots + \frac{1}{n}a_nx^n + c \end{aligned}$$

Note that the degree of the primitive function is one higher than the degree of the gradient function.

The rule for anti-differentiation of a polynomial shows that differentiation and anti-differentiation are inverse operations. For $n \in N$, the derivative of x^n is $n \times x^{n-1}$;

the anti-derivative of x^n is $x^{n+1} \div (n+1) = \frac{x^{n+1}}{n+1} + c$.

WORKED EXAMPLE 1

a If $\frac{dy}{dx} = 6x^8$, find y .

b Find an anti-derivative of $3x^2 - 5x + 2$.

c Given $f'(x) = x(x-1)(x+1)$, find the rule for the primitive function.

THINK

a 1 State the reverse process required to find y .

2 Use the rule to anti-differentiate the polynomial term.

Note: It's a good idea to mentally differentiate the answer to check its derivative is $6x^8$.

b 1 Apply the anti-differentiation rule to each term of the polynomial.

2 Choose any value for the constant c to obtain an anti-derivative.

Note: The conventional choice is $c = 0$.

c 1 Express the rule for the derivative function in expanded polynomial form.

2 Calculate the anti-derivative and state the answer.

WRITE

a $\frac{dy}{dx} = 6x^8$

The derivative of y is $6x^8$. Hence, the anti-derivative of $6x^8$ is y .

$$\begin{aligned} y &= 6 \times \left(\frac{1}{8+1} x^{8+1} \right) + c \\ &= 6 \times \frac{1}{9} x^9 + c \\ &= \frac{2}{3} x^9 + c \end{aligned}$$

b $3x^2 - 5x + 2 = 3x^2 - 5x + 2x^0$

The anti-derivative is:

$$\begin{aligned} &3 \times \frac{1}{3} x^3 - 5 \times \frac{1}{2} x^2 + 2 \times \frac{1}{1} x^1 + c \\ &= x^3 - \frac{5}{2} x^2 + 2x + c \end{aligned}$$

When $c = 0$, an anti-derivative is

$$x^3 - \frac{5}{2} x^2 + 2x.$$

c $f'(x) = x(x-1)(x+1)$

$$= x(x^2 - 1)$$

$$= x^3 - x$$

$$f(x) = \frac{1}{4} x^4 - \frac{1}{2} x^2 + c$$

The rule for the primitive function is

$$f(x) = \frac{1}{4} x^4 - \frac{1}{2} x^2 + c.$$

The indefinite integral

Anti-differentiation is not only about undoing the effect of differentiation. It is an operation that can be applied to functions. There are different forms of notation for the anti-derivative just as there are for the derivative. The most common form of notation uses symbolism due to Leibniz.

It is customary to write 'the anti-derivative of $f(x)$ ' as $\int f(x) dx$. This is called the **indefinite integral**. Using this symbol, we could write 'the anti-derivative of $2x$ with respect to x equals $x^2 + c$ ' as $\int 2x dx = x^2 + c$.

For any function:

$$\int f(x) dx = F(x) + c \text{ where } F'(x) = f(x)$$

$F(x)$ is an anti-derivative or primitive of $f(x)$. It can also be said that $F(x)$ is the indefinite integral, or just integral, of $f(x)$. The arbitrary constant c is called the constant of integration. Here we take integration to be the process of using the integral to obtain an anti-derivative. A little more will be said about integration later. The rule for anti-differentiation of a polynomial term could be written in this notation as:

$$\text{For } n \in \mathbb{N}, \int x^n dx = \frac{x^{n+1}}{n+1} + c$$

The linearity properties of anti-differentiation, or integration, could be expressed as:

$$\int (f(x) \pm g(x)) dx = \int f(x) dx \pm \int g(x) dx$$

$$\int kf(x) dx = k \int f(x) dx$$

Anti-derivatives of power functions

The rule for obtaining the anti-derivative of a polynomial function also holds for power functions where the index may be rational. This is illustrated by applying the rule to $x^{\frac{1}{2}}$.

$$\begin{aligned} \int x^{\frac{1}{2}} dx &= \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} + c \\ &= \frac{x^{\frac{3}{2}}}{\frac{3}{2}} + c \\ &= \frac{2}{3} \times x^{\frac{3}{2}} + c \\ &= \frac{2}{3} x^{\frac{3}{2}} + c \end{aligned}$$

To verify the result, differentiate:

$$\begin{aligned} \frac{d}{dx} \left(\frac{2}{3} x^{\frac{3}{2}} + c \right) &= \frac{2}{3} \times \frac{3}{2} x^{\frac{1}{2}} + 0 \\ &= x^{\frac{1}{2}} \end{aligned}$$

The rule for anti-differentiating x^n is:

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c, n \in \mathbb{R} \setminus \{-1\}$$

This rule must exclude $n = -1$ to prevent the denominator $n + 1$ becoming zero. This exception to the rule will be studied in Mathematical Methods Units 3 and 4.

WORKED EXAMPLE

2

a Use the linearity properties to calculate $\int (2x^3 + 4x - 3) dx$.

b Calculate $\int \frac{7x^5 - 8x^7}{5x} dx$.

c Given $f(x) = \frac{3}{2x^3}$, form $F(x)$, where $F'(x) = f(x)$.

d Find $\int \left(x - \frac{1}{x}\right)^2 dx$.

THINK

a 1 Express the integral as the sum or difference of integrals of each term.

2 Anti-differentiate term by term.

Note: In practice, the calculation usually omits the steps illustrating the linearity properties.

b 1 Express the term to be anti-differentiated in polynomial form.

2 Calculate the anti-derivative.

c 1 Express the given function with a power of x in its numerator.

2 Use the rule to obtain the anti-derivative of the power function.

WRITE

$$\begin{aligned} \mathbf{a} \quad & \int (2x^3 + 4x - 3) dx \\ &= \int 2x^3 dx + \int 4x dx - \int 3 dx \\ &= 2 \int x^3 dx + 4 \int x dx - 3 \int dx \\ &= 2 \times \frac{x^4}{4} + 4 \times \frac{x^2}{2} - 3x + c \\ &= \frac{x^4}{2} + 2x^2 - 3x + c \end{aligned}$$

$$\begin{aligned} \mathbf{b} \quad & \int \frac{7x^5 - 8x^7}{5x} dx \\ &= \int \left(\frac{7x^5}{5x} - \frac{8x^7}{5x} \right) dx \\ &= \int \left(\frac{7}{5}x^4 - \frac{8}{5}x^6 \right) dx \\ &= \frac{7}{5} \times \frac{x^5}{5} - \frac{8}{5} \times \frac{x^7}{7} + c \\ &= \frac{7}{25}x^5 - \frac{8}{35}x^7 + c \end{aligned}$$

$$\begin{aligned} \mathbf{c} \quad & f(x) = \frac{3}{2x^3} \\ &= \frac{3x^{-3}}{2} \end{aligned}$$

$F(x)$ is the anti-derivative function.

$$\begin{aligned} F(x) &= \frac{3}{2} \times \frac{x^{-3+1}}{(-3+1)} + c \\ &= \frac{3}{2} \times \frac{x^{-2}}{-2} + c \\ &= \frac{3x^{-2}}{-4} + c \end{aligned}$$



3 Rewrite the anti-derivative function with positive indices.

$$\therefore F(x) = -\frac{3}{4x^2} + c$$

d 1 Prepare the expression to be anti-differentiated by expanding the perfect square and simplifying.

$$\begin{aligned} \text{d } \int \left(x - \frac{1}{x}\right)^2 dx &= \int \left(x^2 - 2x \times \frac{1}{x} + \frac{1}{x^2}\right) dx \\ &= \int (x^2 - 2 + x^{-2}) dx \\ &= \frac{1}{3}x^3 - 2x + \frac{x^{-2+1}}{-1} + c \\ &= \frac{1}{3}x^3 - 2x - x^{-1} + c \\ &= \frac{1}{3}x^3 - 2x - \frac{1}{x} + c \end{aligned}$$

2 Calculate the anti-derivative.

EXERCISE 14.2 Anti-derivatives

PRACTISE

Work without CAS

- 1 a If $\frac{dy}{dx} = 12x^5$, find y .
 b Find an anti-derivative of $4x^2 + 2x - 5$.
 c Given $f'(x) = (x - 2)(3x + 8)$, obtain the rule for the primitive function.
- 2 Given the gradient function $\frac{dy}{dx} = \frac{2x^3 - 3x^2}{x}$, $x \neq 0$, obtain the primitive function.
- 3 a Use the linearity properties to calculate $\int (-7x^4 + 3x^2 - 6x) dx$.
 b Calculate $\int \frac{5x^8 + 3x^3}{4x^2} dx$.
 c Given $f(x) = \frac{2}{x^4}$, form $F(x)$, where $F'(x) = f(x)$.
 d Obtain $\int \left(x + \frac{1}{x}\right)^2 dx$.
- 4 a Given $f(x) = \frac{1}{2\sqrt{x}}$, obtain $F(x)$, where $F'(x) = f(x)$.
 b Calculate each expression to show that $\int 2(t^2 + 2) dt$ is the same as $2 \int (t^2 + 2) dt$.
- 5 Given $\frac{dy}{dx}$, obtain y in terms of x for each of the following.

a $\frac{dy}{dx} = 5x^9$	b $\frac{dy}{dx} = -3 + 4x^7$
c $\frac{dy}{dx} = 2(x^2 - 6x + 7)$	d $\frac{dy}{dx} = (8 - x)(2x + 5)$

CONSOLIDATE

Apply the most appropriate mathematical processes and tools

6 For each $f'(x)$ expression, obtain an expression for $f(x)$.

a $f'(x) = \frac{1}{2}x^5 + \frac{7}{3}x^6$

b $f'(x) = \frac{4x^6}{3} + 5$

c $f'(x) = \frac{4x^4 - 6x^8}{x^2}, x \neq 0$

d $f'(x) = (3 - 2x^2)^2$

7 Calculate the following indefinite integrals.

a $\int \frac{3x^8}{5} dx$

b $\int 2 dx$

c $4 \int (20x - 5x^7) dx$

d $\int \frac{1}{100} (9 + 6x^2 - 5.5x^{10}) dx$

8 a Calculate the primitive function of $2ax + b$, for $a, b \in R$.

b Calculate the anti-derivative of $0.05x^{99}$.

c Calculate an anti-derivative of $(2x + 1)^3$.

d Calculate the primitive of $7 - x(5x^3 - 4x - 8)$.

9 Find $F(x)$, where $F'(x) = f(x)$ for the following:

a $f(x) = \frac{3x^2 - 2}{4}$

b $f(x) = \frac{3x}{4} + \frac{2(1 - x)}{3}$

c $f(x) = 0.25(1 + 5x^{14})$

d $f(x) = \frac{12(x^5)^2 - (4x)^2}{3x^2}$

10 Given $\frac{dy}{dx}$, obtain y in terms of x for each of the following.

a $\frac{dy}{dx} = x^{\frac{3}{2}}$

b $\frac{dy}{dx} = x^{-\frac{3}{2}}$

11 For each $f'(x)$ expression, obtain an expression for $f(x)$.

a $f'(x) = \frac{5}{x^2}$

b $f'(x) = 6x^3 + 6x^{-3}$

12 Calculate the following indefinite integrals.

a $\int \frac{2x^5 + 7x^3 - 5}{x^2} dx$

b $\int (2\sqrt{x} + 1)^2 dx$

c $\int x^{\frac{1}{5}}(x^2 - 13x^{\frac{1}{10}} + 12) dx$

d $\int \frac{(x + 4)(x - 2)}{2x^4} dx$

13 For the following, find $F(x)$, where $F'(x) = f(x)$, given $a, b \in R$:

a $f(x) = (2ax)^2 + b^3$

b $f(x) = \sqrt{3x^4} + \sqrt{3}a^2$

14 a Calculate the primitive of $(x - 1)(x + 4)(x + 1)$.

b Calculate the anti-derivative of $\left(\frac{5}{x} - \frac{x}{5}\right)^2$.

c Calculate an anti-derivative of $\frac{x^p}{x^q}$, stating the restriction on the values of p and q .

d Calculate an anti-derivative of $\frac{\sqrt[3]{x} - \sqrt[3]{x^5}}{x}$.

e Calculate $\frac{d}{dx} \left(\int (4x + 7) dx \right)$.

15 The calculator enables the anti-derivative to be calculated through a number of menus. Familiarise yourself with these by calculating the anti-derivative of x^3 using each of these methods. Remember to insert the constant yourself if the calculator does not.

16 Write down the answers from the calculator to the following.

a $\int x^{\frac{3}{2}} dx$

b Given $\frac{dy}{dx} = (x^4 + 1)^2$, obtain y .

c If $f(t) = 100\left(t - \frac{5}{t^2}\right)$, obtain $F(t)$.

d $\int (y^5 - y^{-5}) dy$

e $\int \sqrt{4u + 5} du$

f Choose your own function and use CAS technology to obtain its anti-derivative.

14.3

Anti-derivative functions and graphs

If some characteristic information is known about the anti-derivative function, such as a point on its curve, then this information allows the constant of integration to be calculated.

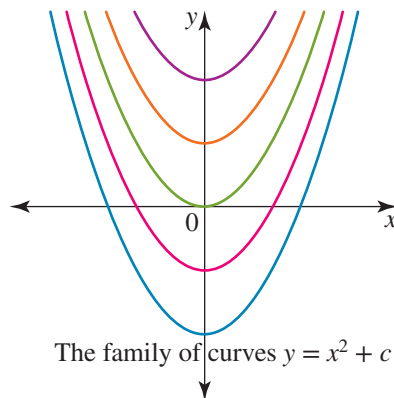
Determining the constant

Once the constant of integration is known, we can determine more than just a family of anti-derivatives. A specific anti-derivative function can be found since only one member of the family will possess the given characteristic information.

While there is a family of anti-derivative functions

$y = x^2 + c$ for which $\frac{dy}{dx} = 2x$, if it is known the

anti-derivative must contain the point $(0, 0)$, substituting $(0, 0)$ in $y = x^2 + c$ gives $0 = 0 + c$ and therefore $c = 0$. Only the specific anti-derivative function $y = x^2$ satisfies this condition.



study on

Units 1 & 2

AOS 3

Topic 3

Concept 2

Anti-derivative functions and graphs

Concept summary
Practice questions

eBookplus

interactivity

Sketching the anti-derivative graph
int-5965

WORKED EXAMPLE 3

a The gradient of a curve is given by $\frac{dy}{dx} = a + 3x$ where a is a constant.

Given the curve has a stationary point at $(2, 5)$, determine its equation.

b If $f'(x) = \frac{x^2 - 4}{2x^2}$ and $f(4) = 3$, obtain the value of $f(1)$.

THINK

a 1 Use the given information to determine the value of a .

WRITE

a $\frac{dy}{dx} = a + 3x$

At the stationary point $(2, 5)$, $\frac{dy}{dx} = 0$.

$\therefore 0 = a + 3 \times 2$

$\therefore a = -6$

2 Calculate the anti-derivative.

$$\frac{dy}{dx} = -6 + 3x$$

$$\begin{aligned}y &= -6x + 3 \times \frac{x^2}{2} + c \\&= -6x + \frac{3x^2}{2} + c\end{aligned}$$

3 Use the given information to calculate the constant of integration.

Substitute the point (2, 5):

$$5 = -6(2) + \frac{3(2)^2}{2} + c$$

$$5 = -12 + 6 + c$$

$$c = 11$$

4 State the answer.

$$y = -6x + \frac{3x^2}{2} + 11$$

The equation of the curve is

$$y = \frac{3}{2}x^2 - 6x + 11.$$

b 1 Express $f'(x)$ in the form in which it can be anti-differentiated.

$$\mathbf{b} \quad f'(x) = \frac{x^2 - 4}{2x^2}$$

$$= \frac{x^2}{2x^2} - \frac{4}{2x^2}$$

$$= \frac{1}{2} - 2x^{-2}$$

2 Calculate the anti-derivative.

$$f(x) = \frac{1}{2}x - \frac{2x^{-1}}{-1} + c$$

$$= \frac{x}{2} + \frac{2}{x} + c$$

3 Use the given information to calculate the constant of integration.

Substitute $f(4) = 3$ to calculate c .

$$3 = \frac{4}{2} + \frac{2}{4} + c$$

$$3 = 2 + \frac{1}{2} + c$$

$$c = \frac{1}{2}$$

4 Calculate the required value.

$$f(x) = \frac{x}{2} + \frac{2}{x} + \frac{1}{2}$$

$$\begin{aligned}f(1) &= \frac{1}{2} + \frac{2}{1} + \frac{1}{2} \\&= 3\end{aligned}$$

Sketching the anti-derivative graph

Given the graph of a function $y = f(x)$ whose rule is not known, the shape of the graph of the anti-derivative function $y = F(x)$ may be able to be deduced from the relationship $F'(x) = f(x)$. This means interpreting the graph of $y = f(x)$ as the gradient graph of $y = F(x)$.

- The x -intercepts of the graph of $y = f(x)$ identify the x -coordinates of the stationary points of $y = F(x)$.
- The nature of any stationary point on $y = F(x)$ is determined by the way the sign of the graph of $y = f(x)$ changes about its x -intercepts.
- If $f(x)$ is a polynomial of degree n then $F(x)$ will be a polynomial of degree $n + 1$.

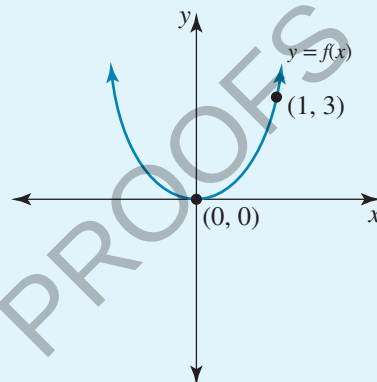
If the rule for $y = f(x)$ can be formed from its graph, then the equation of possible anti-derivative functions can also be formed. Additional information to determine the specific function would be needed, as in the previous Worked example.

WORKED EXAMPLE

4

Consider the graph of the quadratic function $y = f(x)$ shown.

- Describe the position and nature of any stationary point on the graph of $y = F(x)$ where $f(x) = F'(x)$.
- Draw three possible graphs for which $y = F(x)$.
- Obtain the rule for $f(x)$.
- Given $F(0) = -2$, determine the rule for $F(x)$, and sketch the graph of $y = F(x)$.



THINK

- Identify the position of any stationary point on the graph of $y = F(x)$.
 - Determine the nature of the stationary point.
- State the degree of $F(x)$.
 - Draw three possible graphs for the anti-derivative function.

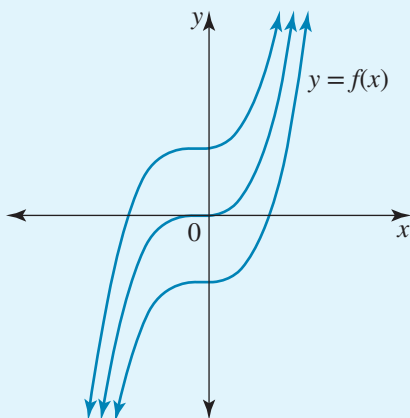
WRITE

- The x -intercept on the given graph identifies a stationary point on $y = F(x)$.
Therefore the graph of $y = F(x)$ has a stationary point at the point where $x = 0$.

As x increases, the sign of the graph of $y = f(x)$ changes from positive to zero to positive about its x -intercept. Therefore, there is a stationary point of inflection at the point where $x = 0$ on the graph of $y = F(x)$.

- Since $f(x)$ is quadratic, its anti-derivative $F(x)$ will be a polynomial of degree 3.

Other than at $x = 0$, the graph of $y = f(x)$ is positive so the gradient of the graph of $y = F(x)$ is positive other than at $x = 0$. The y -coordinate of the stationary point on $y = F(x)$ is not known. Three graphs of an increasing cubic function with a stationary point of inflection at the point where $x = 0$ are shown for $y = F(x)$.



- c** Determine the equation of the quadratic function using the information on its graph.

- c** The graph of $y = f(x)$ has a minimum turning point at $(0, 0)$ and contains the point $(1, 3)$.

$$\text{Let } f(x) = a(x - h)^2 + k.$$

The turning point is $(0, 0)$.

$$\therefore f(x) = ax^2$$

Substitute the point $(1, 3)$:

$$3 = a(1)^2$$

$$\therefore a = 3$$

The rule for the quadratic function is $f(x) = 3x^2$.

- d 1** Use anti-differentiation to obtain the rule for $F(x)$.

d $f(x) = 3x^2$
 $\therefore F(x) = x^3 + c$

$$\text{Since } F(0) = -2,$$

$$-2 = (0)^3 + c$$

$$c = -2$$

$$\therefore F(x) = x^3 - 2$$

- 2** Determine the value of the constant of integration using the given information and state the rule for $F(x)$.

- 3** Sketch the graph of $y = F(x)$.

The point $(0, -2)$ is a stationary point of inflection and the y -intercept.

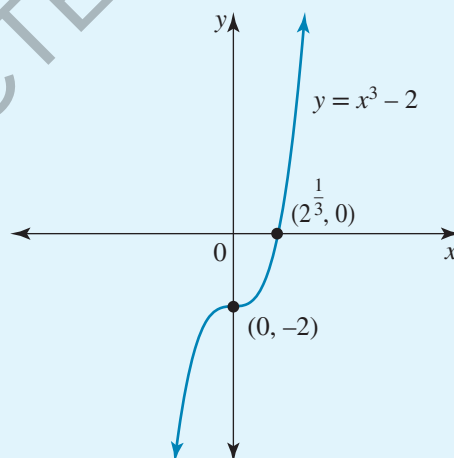
x -intercept: let $y = 0$

$$x^3 - 2 = 0$$

$$x^3 = 2$$

$$x = \sqrt[3]{2}$$

$(\sqrt[3]{2}, 0)$ is the x -intercept.



EXERCISE 14.3 Anti-derivative functions and graphs

PRACTISE

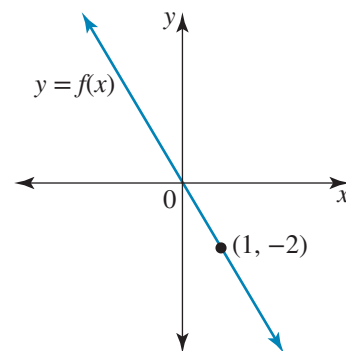
Work without CAS

- 1 a** The gradient of a curve is given by $\frac{dy}{dx} = ax - 6$ where a is a constant. Given the curve has a stationary point at $(-1, 10)$, determine its equation.
- b** If $f'(x) = \frac{2x^2 + 9}{2x^2}$ and $f(3) = 0$, find the value of $f(-1)$.
- 2** If $\frac{dy}{dx} = 2\sqrt{x}$ and $y = 10$ when $x = 4$, find y when $x = 1$.

CONSOLIDATE

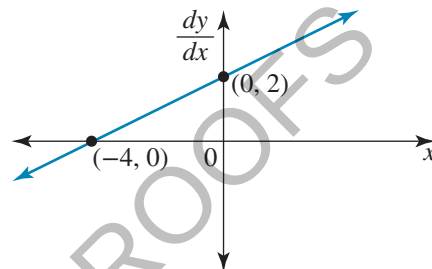
Apply the most appropriate mathematical processes and tools

- 3 Consider the graph of the linear function $y = f(x)$ shown.
 - a Describe the position and nature of any stationary point on the graph of $y = F(x)$ where $f(x) = F'(x)$.
 - b Draw three possible graphs for which $y = F(x)$.
 - c Obtain the rule for $f(x)$.
 - d Given $F(0) = 1$, determine the rule for $F(x)$ and sketch the graph of $y = F(x)$.



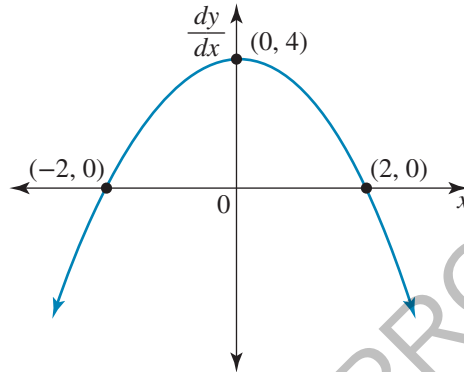
- 4 The graph of the gradient, $\frac{dy}{dx}$, of a particular curve is shown.

Given that $(-2, 3)$ lies on the curve with this gradient, determine the equation of the curve and deduce the coordinates of, and the nature of, any stationary point on the curve.

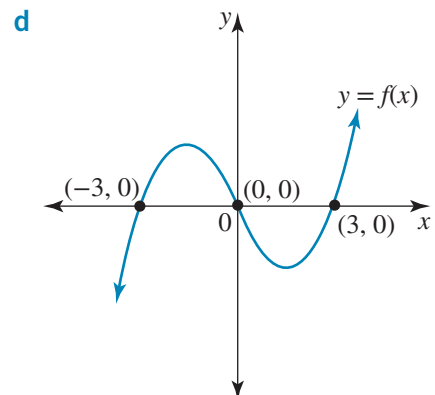
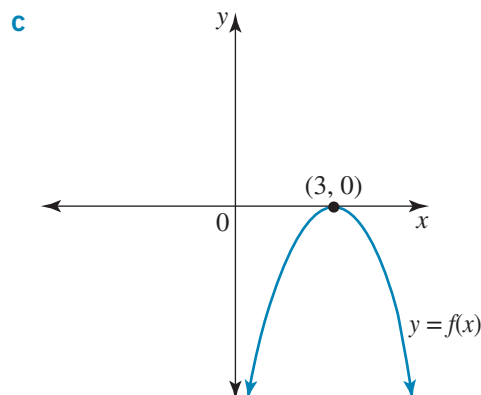
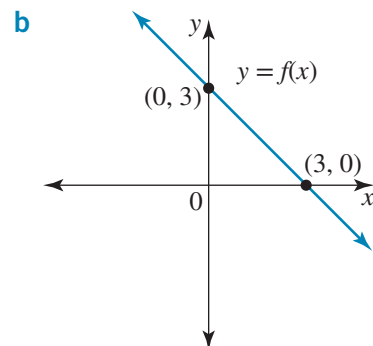
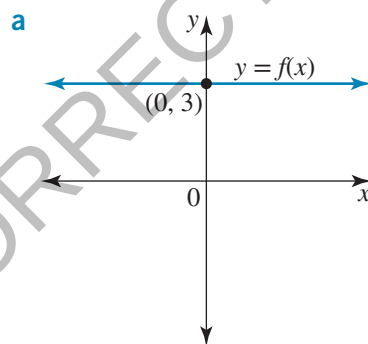


- 5 The gradient of a curve is given by $f'(x) = -3x^2 + 4$. Find the equation of the curve if it passes through the point $(-1, 2)$.
- 6 The gradient of a curve is given by $\frac{dy}{dx} = \frac{2x}{5} - 3$. It is also known that the curve passes through the point $(5, 0)$.
 - a Determine the equation of the curve.
 - b Find its x -intercepts.
- 7 The gradient of a curve is directly proportional to x and at the point $(2, 5)$ on the curve the gradient is -3 .
 - a Determine the constant of proportionality.
 - b Find the equation of the curve.
- 8 A function is defined by $f(x) = \frac{(4-x)(5-x)}{10x^4}$. Determine its primitive function if the point $(1, -1)$ lies on the primitive function.
- 9 a Given $\frac{dy}{dx} = 2x(3-x)$ and $y = 0$ when $x = 3$, obtain the value of y when $x = 0$.
 - b Given $\frac{dz}{dx} = (10-x)^2$ and $z = 200$ when $x = 10$, obtain the value of z when $x = 4$.
 - c Given $\frac{dA}{dt} = \frac{4}{\sqrt{t}}$ and $A = 40$ when $t = 16$, obtain the value of A when $t = 64$.
 - d Given $\frac{dx}{dy} = -\frac{3}{y^4}$ and $y = 1$ when $x = 12$, obtain the value of y when $x = 75$.
- 10 At any point (x, y) on a curve, the gradient of the tangent to the curve is given by $\frac{dy}{dx} = a - x^{\frac{2}{3}}$. The curve has a stationary point at $(8, 32)$.
 - a Find the value of a .
 - b Determine the equation of the curve.
 - c Calculate the equation of the tangent to the curve at the point where $x = 1$.

- 11** A curve with a horizontal asymptote with equation $y = a$ passes through the point $(2, 3)$. If the gradient of the curve is given by $F'(x) = \frac{a}{x^2}$:
- Find the value of a .
 - Determine the equation of the curve.
 - Calculate, to the nearest degree, the angle at which the curve cuts the x -axis.
- 12** The gradient function of a curve is illustrated in the diagram.



- Describe the position and nature of the stationary points of the curve with this gradient function.
 - Sketch a possible curve which has this gradient function.
 - Determine the rule for the gradient graph and hence obtain the equation which describes the family of curves with this gradient function.
 - One of the curves belonging to the family of curves cuts the x -axis at $x = 3$. Obtain the equation of this particular curve and calculate the slope with which it cuts the x -axis at $x = 3$.
- 13** For each of the following graphs of $y = f(x)$, draw a sketch of a possible curve for $y = F(x)$ given $F'(x) = f(x)$.

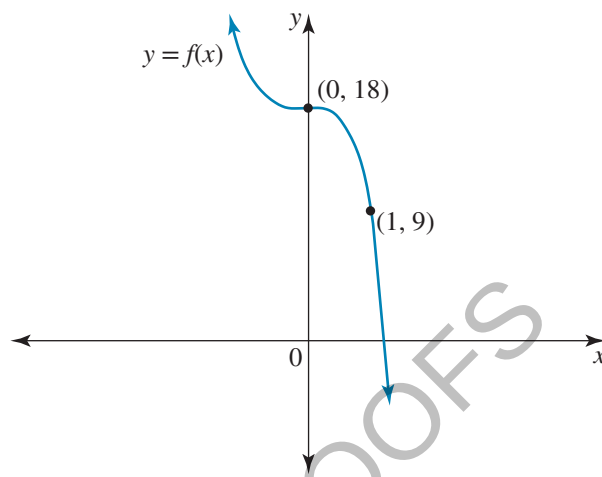


- 14 The diagram shows the graph of a cubic function $y = f(x)$ with a stationary point of inflection at $(0, 18)$ and passing through the point $(1, 9)$.

a Find $\int f(x) dx$.

b A particular anti-derivative function of $y = f(x)$ passes through the origin. For this particular function:

- Determine its equation.
- Find the coordinates of its other x -intercept and determine at which of its x -intercepts the graph of this function is steeper.
- Deduce the exact x -coordinate of its turning point and its nature.
- Hence show the y -coordinate of its turning point can be expressed as $2^p \times 3^q$, specifying the values of p and q .
- Draw a sketch graph of this particular anti-derivative function.



MASTER

- 15 Use the CAS calculator's keyboard template to sketch the shape of the graphs of the anti-derivative functions given by:

a $y_1 = \int (x - 2)(x + 1) dx$

b $y_2 = \int (x - 2)^2(x + 1) dx$

c $y_3 = \int (x - 2)^2(x + 1)^2 dx$.

d Comment on the effect of changing the multiplicity of each factor.

- 16 Define $f(x) = x^2 - 6x$ and use the calculator to sketch on the same set of axes the graphs of $y = f(x)$, $y = f'(x)$ and $y = \int f(x) dx$. Comment on the connections between the three graphs.

14.4 Applications of anti-differentiation

In this section, anti-differentiation is applied in calculations involving functions defined by their rates of change.

study on

Units 1 & 2

AOS 3

Topic 3

Concept 3

Applications of anti-differentiation

Concept summary
Practice questions

The anti-derivative in kinematics

Recalling that velocity, v , is the rate of change of displacement, x , means that we can now interpret displacement as the anti-derivative of velocity.

$$v = \frac{dx}{dt} \Leftrightarrow x = \int v dt$$

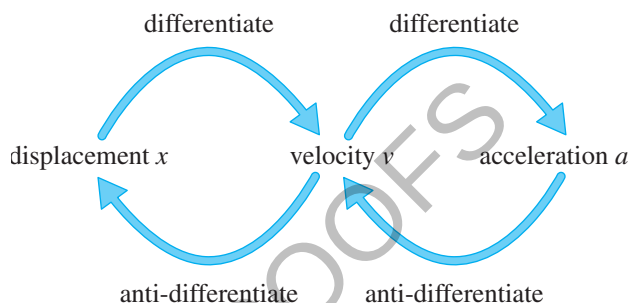
Further, since acceleration is the rate of change of velocity, it follows that velocity is the anti-derivative of acceleration.

$$a = \frac{dv}{dt} \Leftrightarrow v = \int a dt$$

The role of differentiation and anti-differentiation in kinematics is displayed in the diagram below.

Velocity is of particular interest for two reasons:

- The anti-derivative of velocity with respect to time gives displacement.
- The derivative of velocity with respect to time gives acceleration.



WORKED EXAMPLE

5

A particle moves in a straight line so that its velocity at time t seconds is given by $v = 3t^2 + 4t - 4$, $t \geq 0$. Initially the particle is 8 metres to the left of a fixed origin.

- After how many seconds does the particle reach the origin?
- Calculate the particle's acceleration when its velocity is zero.

THINK

- Calculate the displacement function from the velocity function.
- Use the initial conditions to calculate the constant of integration.
- Set up and solve the equation which gives the time the particle is at the required position.

WRITE

a $v = 3t^2 + 4t - 4$

Anti-differentiate the velocity to obtain displacement.

$$\begin{aligned} x &= \frac{3t^3}{3} + \frac{4t^2}{2} - 4t + c \\ &= t^3 + 2t^2 - 4t + c \end{aligned}$$

When $t = 0$, $x = -8$.

$$\therefore -8 = c$$

$$\therefore x = t^3 + 2t^2 - 4t - 8$$

When the particle is at the origin, $x = 0$.

$$t^3 + 2t^2 - 4t - 8 = 0$$

$$\therefore t^2(t + 2) - 4(t + 2) = 0$$

$$\therefore (t + 2)(t^2 - 4) = 0$$

$$\therefore (t + 2)^2(t - 2) = 0$$

$$\therefore t = -2, t = 2t$$

As $t \geq 0$, reject $t = -2$.

$$\therefore t = 2$$

The particle reaches the origin after 2 seconds.



b 1 Calculate the acceleration function from the velocity function.

2 Calculate the time when the velocity is zero.

3 Calculate the acceleration at the time required.

$$v = 3t^2 + 4t - 4$$

$$\text{Acceleration } a = \frac{dv}{dt}$$

$$\therefore a = 6t + 4.$$

When velocity is zero,

$$3t^2 + 4t - 4 = 0$$

$$\therefore (3t - 2)(t + 2) = 0$$

$$\therefore t = \frac{2}{3}, t = -2$$

$$t \geq 0, \text{ so } t = \frac{2}{3}$$

$$\text{When } t = \frac{2}{3},$$

$$a = 6 \times \frac{2}{3} + 4 \\ = 8$$

The acceleration is 8 m/s^2 when the velocity is zero.

Acceleration and displacement

To obtain the displacement from an expression for acceleration will require anti-differentiating twice in order to proceed from a to v to x . The first anti-differentiation operation will give velocity and the second will obtain displacement from velocity.

This will introduce two constants of integration, one for each step. Where possible, evaluate the first constant using given $v - t$ information before commencing the second anti-differentiation operation. To evaluate the second constant, $x - t$ information will be needed. The notation used for each constant should distinguish between them. For example, the first constant could be written as c_1 and the second as c_2 .

WORKED
EXAMPLE

6

A particle moves in a straight line so that its acceleration at time t seconds is given by $a = 4 + 6t$, $t \geq 0$. If $v = 30$ and $x = 2$ when $t = 0$, calculate the particle's velocity and position when $t = 1$.

THINK

- 1 Calculate the velocity function from the acceleration function.
- 2 Evaluate the first constant of integration using the given $v - t$ information.
- 3 Calculate the displacement function from the velocity function.
- 4 Evaluate the second constant of integration using the given $x - t$ information.

WRITE

$$a = 4 + 6t$$

Anti-differentiate the acceleration to obtain velocity.

$$v = 4t + 3t^2 + c_1 \\ = 3t^2 + 4t + c_1$$

$$\text{When } t = 0, v = 30$$

$$\Rightarrow 30 = c_1$$

$$\therefore v = 3t^2 + 4t + 30$$

Anti-differentiate the velocity to obtain displacement.

$$x = t^3 + 2t^2 + 30t + c_2$$

$$\text{When } t = 0, x = 2$$

$$\Rightarrow 2 = c_2$$

$$\therefore x = t^3 + 2t^2 + 30t + 2$$

5 Calculate the velocity at the given time.

$$v = 3t^2 + 4t + 30$$

When $t = 1$,

$$v = 3 + 4 + 30$$

$$= 37$$

6 Calculate the displacement at the given time.

$$x = t^3 + 2t^2 + 30t + 2$$

When $t = 1$,

$$x = 1 + 2 + 30 + 2$$

$$= 35$$

7 State the answer.

The particle has a velocity of 37 m/s and its position is 35 metres to the right of the origin when $t = 1$.

Other rates of change

The process of anti-differentiation can be used to solve problems involving other rates of change, not only those involved in kinematics. For example, if the rate at which the area of an oil spill is changing with respect to t is given by $\frac{dA}{dt} = f(t)$, then the area of the oil spill at time t is given by $A = \int f(t) dt$, the anti-derivative with respect to t .

WORKED
EXAMPLE

7

An ice block with initial volume $36\pi \text{ cm}^3$ starts to melt. The rate of change of its volume $V \text{ cm}^3$ after t seconds is given by $\frac{dV}{dt} = -0.2$.

a Use anti-differentiation to express V in terms of t .

b Calculate, to one decimal place, the number of minutes it takes for all of the ice block to melt.

THINK

a 1 Calculate the volume function from the given derivative function.

2 Use the initial conditions to evaluate the constant of integration.

b 1 Identify the value of V and calculate the corresponding value of t .

2 Express the time in the required units and state the answer to the required degree of accuracy.

Note: The rate of decrease of the volume is constant, so this problem could have been solved without calculus.

WRITE

a $\frac{dV}{dt} = -0.2$

Anti-differentiate with respect to t :

$$V = -0.2t + c$$

$$\text{When } t = 0, V = 36\pi$$

$$\therefore 36\pi = c$$

$$\text{Therefore, } V = -0.2t + 36\pi$$

b When the ice block has melted, $V = 0$.

$$0 = -0.2t + 36\pi$$

$$\therefore t = \frac{36\pi}{0.2}$$

$$= 180\pi$$

The ice block melts after 180π seconds.

Divide this by 60 to convert to minutes:

180π seconds is equal to 3π minutes.

To one decimal place, the time for the ice block to melt is 9.4 minutes.

PRACTISE

Work without CAS

- A particle moves in a straight line so that its velocity at time t seconds is given by $v = 3t^2 - 10t - 8$, $t \geq 0$. Initially the particle is 40 metres to the right of a fixed origin.
 - After how many seconds does the particle first reach the origin?
 - Calculate the particle's acceleration when its velocity is zero.
- A particle moves in a straight line so that its velocity at time t seconds is given by $v = \frac{1}{t^2} + 2$, $t > 0$. When $t = 1$, the particle is 1 metre to the left of a fixed origin. Obtain expressions for the particle's displacement and acceleration after t seconds.
- A particle moves in a straight line so that its acceleration at time t seconds is given by $a = 8 - 18t$, $t \geq 0$. If $v = 10$ and $x = -2$ when $t = 0$, calculate the particle's velocity and position when $t = 1$.
- A particle moves in a straight line so that its acceleration, in m/s^2 , at time t seconds is given by $a = 9.8$, $t \geq 0$. If $v = x = 0$ when $t = 0$, calculate the particle's displacement when $t = 5$.
- An ice block with initial volume $4.5\pi \text{ cm}^3$ starts to melt. The rate of change of its volume, $V \text{ cm}^3$, after t seconds is given by $\frac{dV}{dt} = -0.25$.
 - Use anti-differentiation to express V in terms of t .
 - Calculate, to one decimal place, the number of seconds it takes for all of the ice block to melt.
- When first purchased, the height of a small rubber plant was 50 cm. The rate of growth of its height over the first year after its planting is measured by $h'(t) = 0.2t$, $0 \leq t \leq 12$, where h is its height in cm t months after being planted. Calculate its height at the end of the first year after its planting.
- The velocity, $v \text{ m/s}$, of a particle moving in a straight line at time t seconds is given by $v = 8t^2 - 20t - 12$, $t \geq 0$. Initially the particle is 54 metres to the right of a fixed origin.
 - Obtain an expression for the particle's displacement at time t seconds.
 - How far from its initial position is the particle after the first second?
 - Determine the position of the particle when its velocity is zero.
- A particle starts from rest and moves in a straight line so that its velocity at time t is given by $v = -3t^3$, $t \geq 0$. When its position is 1 metre to the right of a fixed origin, its velocity is -24 m/s . What was the initial position of the particle relative to the fixed origin?
- Starting from a point 9 metres to the right of a fixed origin, a particle moves in a straight line in such a way that its velocity after t seconds is $v = 6 - 6t$, $t \geq 0$.
 - Show that the particle moves with constant acceleration.
 - Determine when and where its velocity is zero.
 - How far does the particle travel before it reaches the origin?
 - What is the average speed of this particle over the first three seconds?
 - What is the average velocity of this particle over the first three seconds?

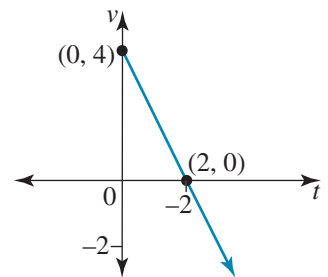


CONSOLIDATE

Apply the most appropriate mathematical processes and tools

- 10 After t seconds the velocity v m/s of a particle moving in a straight line is given by $v(t) = 3(t - 2)(t - 4)$, $t \geq 0$.
- Calculate the particle's initial velocity and initial acceleration.
 - Obtain an expression for $x(t)$, the particle's displacement from a fixed origin after time t seconds, given that the particle was initially at this origin.
 - Calculate $x(5)$.
 - How far does the particle travel during the first 5 seconds?
- 11 A particle moving in a straight line has a velocity of 3 m/s and a displacement of 2 metres from a fixed origin after 1 second. If its acceleration after time t seconds is $a = 8 + 6t$, obtain expressions in terms of t for:
- its velocity
 - its displacement.
- 12 The acceleration of a moving object is given by $a = -10$. Initially, the object was at the origin and its initial velocity was 20 m/s.
- Determine when and where its velocity is zero.
 - After how many seconds will the object return to its starting point?

- 13 The diagram shows a velocity–time graph.



- Draw a possible displacement–time graph.
- Draw the acceleration–time graph.
- If it is known that the initial displacement is -4 , determine the rule for the displacement–time graph.
- Sketch the displacement–time graph with the rule obtained in part c.

- 14 Pouring boiling water on weeds is a means of keeping unwanted weeds under control. Along the cracks in an asphalt driveway, a weed had grown to cover an area of 90 cm^2 , so it was given the boiling water treatment.



The rate of change of the area, $A \text{ cm}^2$, of the weed t days after the boiling water is poured on it is given by $\frac{dA}{dt} = -18t$.

- Express the area as a function of t .
- According to this model, how many whole days will it take for the weed to be completely removed?
- For what exact values of t is the model $\frac{dA}{dt} = -18t$ valid?

- 15 Rainwater collects in a puddle on part of the surface of an uneven path. For the time period for which the rain falls, the rate at which the area, $A \text{ m}^2$, grows is modelled by $A'(t) = 4 - t$, where t is the number of days of rain.



- After how many days does the area of the puddle stop increasing?
- Express the area function $A(t)$ in terms of t and state its domain.
- What is the greatest area the puddle grows to?
- On the same set of axes, sketch the graphs of $A'(t)$ versus t and $A(t)$ versus t over an appropriate domain.

- 16 The rate of growth of a colony of microbes in a laboratory can be modelled by $\frac{dm}{dt} = k\sqrt{t}$ where m is the number of microbes after t days. Initially there were 20 microbes and after 4 days the population was growing at 300 microbes per day.
- Determine the value of k .
 - Express m as a function of t .
 - Hence calculate the number of days it takes for the population size to reach 6420.

MASTER

- 17 The velocity of an object which moves in a straight line is $v = (2t + 1)^4$, $t \geq 0$. Use CAS technology to:
- state the acceleration
 - calculate the displacement, given that initially the object's displacement was 4.2 metres from a fixed origin
 - find the time and the velocity, to two decimal places, when the displacement is 8.4 metres.
- 18 With the aid of CAS technology, obtain an expression for x in terms of t if $a = \frac{1}{(t + 1)^3}$ and $x = v = 0$ when $t = 0$.

14.5 The definite integral

study on

Units 1 & 2

AOS 3

Topic 3

Concept 4

The definite integral

Concept summary
Practice questions

The indefinite integral has been used as a symbol for the anti-derivative. In this section we will look at the definite integral. We shall learn how to evaluate a definite integral, but only briefly explore its meaning, since this will form part of the Mathematical Methods Units 3 and 4 course.

The definite integral

The definite integral is of the form $\int_a^b f(x) dx$; it is quite similar to the indefinite

integral but has the numbers a and b placed on the integral symbol. These numbers are called terminals. They define the endpoints of the interval or the limits on the values of the variable x over which the integration takes place. Due to their relative positions, a is referred to as the **lower terminal** and b as the **upper terminal**.

For example, $\int_1^3 2x dx$ is a definite integral with lower terminal 1 and upper terminal 3,

whereas $\int 2x dx$ without terminals is an indefinite integral.

In the definite integral the term $f(x)$ is called the **integrand**. It is the expression being integrated with respect to x .

Calculation of the definite integral

The definite integral results in a numerical value when it is evaluated.

It is evaluated by the calculation: $\int_a^b f(x) dx = F(b) - F(a)$, where $F(x)$ is an anti-derivative of $f(x)$.

The calculation takes two steps:

- Anti-differentiate $f(x)$ to obtain $F(x)$.
- Substitute the terminals in $F(x)$ and carry out the subtraction calculation.

This calculation is commonly written as:

$$\begin{aligned}\int_a^b f(x) dx &= [F(x)]_a^b \\ &= F(b) - F(a)\end{aligned}$$

For example, to evaluate $\int_1^3 2x dx$, we write:

$$\begin{aligned}\int_1^3 2x dx &= [x^2]_1^3 \\ &= (3)^2 - (1)^2 \\ &= 8\end{aligned}$$

Note that only 'an' anti-derivative is needed in the calculation of the definite integral. Had the constant of integration been included in the calculation, these terms would cancel out, as illustrated by the following:

$$\begin{aligned}\int_1^3 2x dx &= [x^2 + c]_1^3 \\ &= ((3)^2 + c) - ((1)^2 + c) \\ &= 9 + c - 1 - c \\ &= 8\end{aligned}$$

The value of a definite integral may be a positive, zero or negative real number. It is not a family of functions as obtained from an indefinite integral.

WORKED EXAMPLE 8

Evaluate $\int_{-1}^2 (3x^2 + 1) dx$.

THINK

- 1 Calculate an anti-derivative of the integrand.
- 2 Substitute the upper and then the lower terminal in place of x and subtract the two expressions.
- 3 Evaluate the expression.
- 4 State the answer.

WRITE

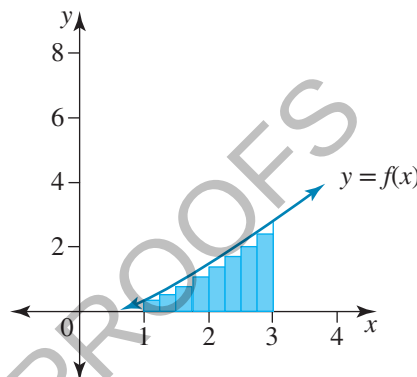
$$\begin{aligned}\int_{-1}^2 (3x^2 + 1) dx &= [x^3 + x]_{-1}^2 \\ &= (2^3 + 2) - ((-1)^3 + (-1)) \\ &= (8 + 2) - (-1 - 1) \\ &= (10) - (-2) \\ &= 12 \\ \int_{-1}^2 (3x^2 + 1) dx &= 12\end{aligned}$$

Integration and area

Areas of geometric shapes such as rectangles, triangles, and trapeziums can be used to approximate areas enclosed by edges which are not straight. Leibniz invented the method for calculating the exact measure of an area enclosed by a curve, thereby establishing the branch of calculus known as integral calculus. His method bears some similarity to a method used in the geometry studies of Archimedes around 230 BC.

Leibniz's approach involved summing a large number of very small areas and then calculating the limiting sum, a process known as **integration**. Like differential calculus, integral calculus is based on the concept of a limit.

To calculate the area bounded by a curve $y = f(x)$ and the x -axis between $x = 1$ and $x = 3$, the area could be approximated by a set of rectangles as illustrated in the diagram below. Here, 8 rectangles, each of width 0.25 units, have been constructed.



The larger the number of rectangles, the better the accuracy of the approximation to the actual area; also, the larger the number, the smaller the widths of the rectangles.

To calculate the area between $x = a$ and $x = b$, Leibniz partitioned (divided) the interval $[a, b]$ into a large number of strips of very small width. A typical strip would have width δx and area δA , with the total area approximated by the sum of the areas of such strips.

For δx , the approximate area of a typical strip is $\delta A \approx y\delta x$, so the total area A is approximated from the sum of these areas. This is written as

$$A \approx \sum_{x=a}^b y \delta x.$$

The approximation improves as $\delta x \rightarrow 0$ and the number of strips increases.

The actual area is calculated as a limit; the limiting sum as $\delta x \rightarrow 0$, and

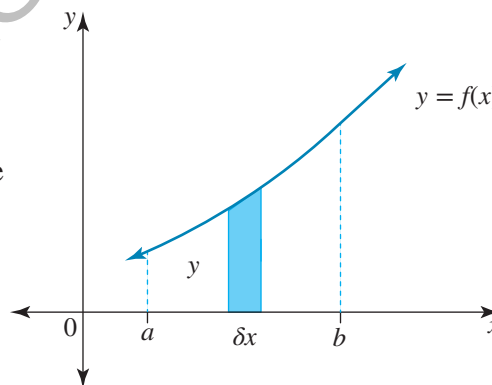
$$\text{therefore } A = \lim_{\delta x \rightarrow 0} \sum_{x=a}^b y \delta x.$$

Leibniz chose to write the symbol for the limiting sum with the long capital 'S' used in his time. That symbol is the now familiar integral sign. This means that the area

bounded by the curve, the x -axis and $x = a, x = b$ is $A = \int_a^b y dx$ or $A = \int_a^b f(x) dx$.

The process of evaluating this integral or limiting sum expression is called integration.

The fundamental theorem of calculus links together the process of anti-differentiation with the process of integration. The full proof of this theorem is left to Units 3 and 4.



Signed area

As we have seen, the definite integral can result in a positive, negative or zero value. Area, however, can only be positive.

If $f(x) > 0$, the graph of $y = f(x)$ lies above the x -axis and $\int_a^b f(x) dx$ will be positive.

Its value will be a measure of the area bounded by the curve $y = f(x)$ and the x -axis between $x = a$ and $x = b$.

However, if $f(x) < 0$, the graph of $y = f(x)$ is below the x -axis and $\int_a^b f(x) dx$ will be negative. Its value gives a **signed area**. By ignoring the negative sign, the value of the integral will still measure the actual area bounded by the curve $y = f(x)$ and the x -axis between $x = a$ and $x = b$.

If over the interval $[a, b]$ $f(x)$ is partly positive and partly negative, then $\int_a^b f(x) dx$ could be positive or negative, or even zero, depending on which of the positive or negative signed areas is the 'larger'.

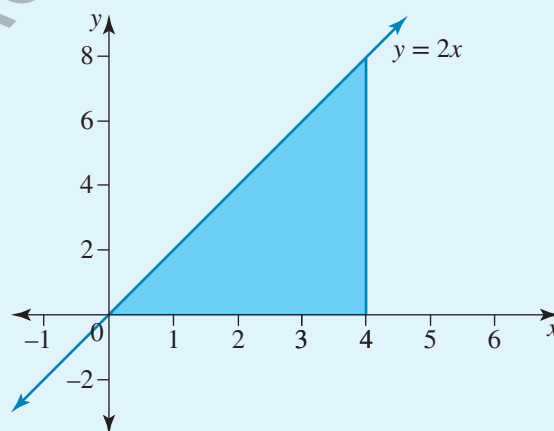
$\int_a^b f(x) dx$ is the sum of signed area measures

This adds a complexity to using the signed area measure of the definite integral to calculate area. Perhaps you can already think of a way around this situation. However, for now, another treat awaiting you in Units 3 and 4 will be to explore signed areas.

WORKED EXAMPLE 9

The area bounded by the line $y = 2x$, the x -axis and $x = 0$, $x = 4$ is illustrated in the diagram.

- Calculate the area using the formula for the area of a triangle.
- Write down the definite integral which represents the measure of this area.
- Hence, calculate the area using calculus.



THINK

- Calculate the area using the formula for the area of a triangle.

- State the definite integral which gives the area.

WRITE

- Base is 4 units; height is 8 units.

Area of triangle:

$$A = \frac{1}{2}bh$$

$$= \frac{1}{2} \times 4 \times 8$$

$$= 16$$

The area is 16 square units.

- The definite integral $\int_a^b f(x) dx$ gives the area. For this area $f(x) = 2x$, $a = 0$ and $b = 4$.

$$\int_0^4 2x dx \text{ gives the area measure.}$$



- c 1** Evaluate the definite integral.

Note: There are no units, just a real number for the value of a definite integral.

$$\begin{aligned} \text{c } \int_0^4 2x dx &= [x^2]_0^4 \\ &= 4^2 - 0^2 \\ &= 16 \end{aligned}$$

- 2** State the area using appropriate units.

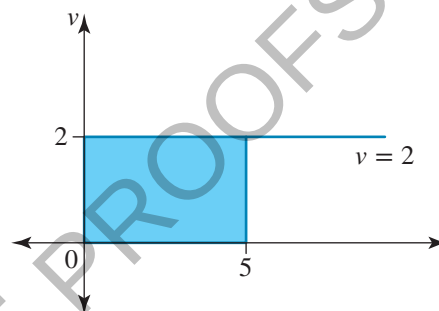
Therefore, the area is 16 square units.

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int-5964

The definite integral in kinematics

A particle which travels at a constant velocity of 2 m/s will travel a distance of 10 metres in 5 seconds. For this motion, the velocity–time graph is a horizontal line. Looking at the rectangular area under the velocity–time graph bounded by the horizontal axis between $t = 0$ and $t = 5$ shows that its area measure is also equal to 10.



- The area under the velocity–time graph gives the measure of the distance travelled by a particle.
- For a positive signed area, the definite integral $\int_{t_1}^{t_2} v dt$ gives the distance travelled over the time interval $t \in [t_1, t_2]$.

The velocity–time graph is the most important of the motion graphs as it gives the velocity at any time; the gradient of its tangent gives the instantaneous acceleration and the area under its graph gives the distance travelled.

WORKED EXAMPLE 10

The velocity of a particle moving in a straight line is given by $v = 3t^2 + 1$, $t \geq 0$.

- Give the particle's velocity and acceleration when $t = 1$.
- Sketch the velocity–time graph and shade the area which represents the distance the particle travels over the interval $t \in [1, 3]$.
 - Use a definite integral to calculate the distance.

Assume units for distance are in metres and time in seconds.

THINK

- a 1** Calculate the velocity at the given time.

WRITE

$$\begin{aligned} \text{a } v &= 3t^2 + 1 \\ \text{When } t &= 1, \\ v &= 3 \times 1^2 + 1 \\ &= 4 \\ \text{The velocity is } &4 \text{ m/s.} \end{aligned}$$

- 2** Obtain the acceleration from the velocity function and evaluate it at the given time.

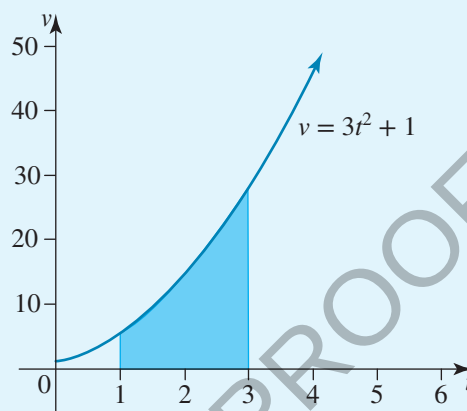
$$\begin{aligned} a &= \frac{dv}{dt} \\ &= 6t \\ \text{When } t &= 1, a = 6 \\ \text{The acceleration is } &6 \text{ m/s}^2. \end{aligned}$$

- b i** Sketch the $v-t$ graph and shade the area required.

b i $v = 3t^2 + 1, t \geq 0$

The $v-t$ graph is part of the parabola which has turning point $(0, 1)$ and passes through the point $(1, 4)$.

The area under the graph bounded by the t -axis and $t = 1, t = 3$ represents the distance.



- ii 1** State the definite integral which gives the measure of the area shaded.

Note: The integrand is a function of t and is to be integrated with respect to t .

ii Area measure is given by $\int_1^3 (3t^2 + 1) dt$.

- 2** Evaluate the definite integral.

$$\begin{aligned} \int_1^3 (3t^2 + 1) dt &= [t^3 + t]_1^3 \\ &= (3^3 + 3) - (1^3 + 1) \\ &= 28 \end{aligned}$$

- 3** State the distance travelled.

The distance travelled by the particle over the time interval $[1, 3]$ is 28 metres.

EXERCISE 14.5 The definite integral

PRACTISE

Work without CAS

1 Evaluate $\int_0^3 (3x^2 - 2x) dx$.

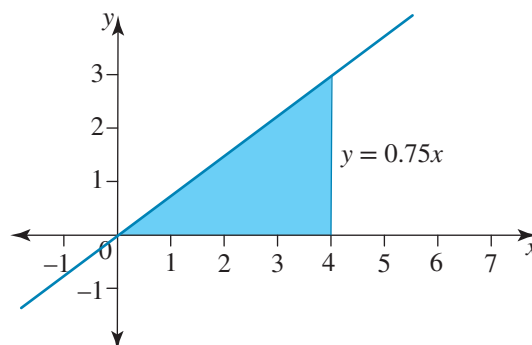
- 2** Evaluate the following.

a $\int_{-2}^2 (x - 2)(x + 2) dx$

b $\int_{-1}^1 x^3 dx$

- 3** The area bounded by the line $y = 0.75x$, the x -axis and $x = 0, x = 4$ is illustrated in the diagram.

- a** Calculate the area using the formula for the area of a triangle.



- b** Write down the definite integral which represents the measure of this area.
c Hence, calculate the area using calculus.

- 4** The area bounded by the curve $y = 16 - x^2$, the x -axis and $x = 1$, $x = 3$ is illustrated in the diagram.

- a** Write down the definite integral which represents the measure of this area.
b Hence, calculate the area.

- 5** The velocity of a particle moving in a straight line is given by $v = t^3 + 2$, $t \geq 0$.

- a** Give the particle's velocity and acceleration when $t = 1$.
b i Sketch the velocity–time graph and shade the area which represents the distance the particle travels over the interval $t \in [2, 4]$.
ii Use a definite integral to calculate the distance.
 Assume units for distance are in metres and time is in seconds.

- 6** The velocity, v m/s, of a particle moving in a straight line after t seconds is given by $v = 3t^2 - 2t + 5$, $t \geq 0$.

- a** Show that velocity is always positive.
b Calculate the distance the particle travels in the first 2 seconds using:
i a definite integral
ii anti-differentiation.

- 7** Evaluate each of the following.

a $\int_{-2}^2 5x^4 dx$

c $\int_1^3 (6x^2 + 5x - 1) dx$

e $\int_{-4}^{-2} (x + 4)^2 dx$

b $\int_0^2 (7 - 2x^3) dx$

d $\int_{-3}^0 12x^2(x - 1) dx$

f $\int_{-\frac{1}{2}}^{\frac{1}{2}} (x + 1)(x^2 - x) dx$

- 8** Show by calculation that the following statements are true.

a $\int_1^2 (20x + 15) dx = 5 \int_1^2 (4x + 3) dx$

c $\int_1^3 3x^2 dx = \int_1^3 3t^2 dt$

e $\int_b^a 3x^2 dx = - \int_a^b 3x^2 dx$

b $\int_{-1}^2 (x^2 + 2) dx = \int_{-1}^2 x^2 dx + \int_{-1}^2 2 dx$

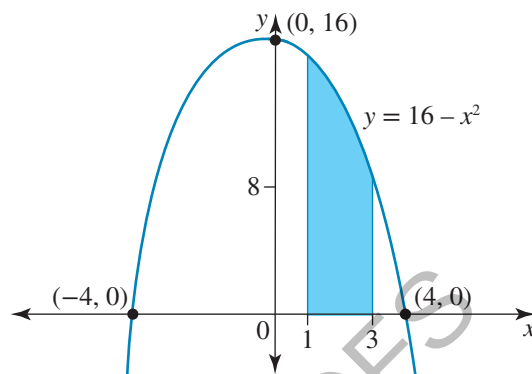
d $\int_a^a 3x^2 dx = 0$

f $\int_{-a}^a dx = 2a$

- 9** Evaluate each of the following.

a $\int_1^2 \frac{1}{x^2} dx$

b $\int_0^9 \sqrt{x} dx$



CONSOLIDATE

Apply the most appropriate mathematical processes and tools

$$\text{c } \int_1^4 5x^{\frac{3}{2}} dx$$

$$\text{d } \int_1^2 \frac{x^6 + 2x^3 + 4}{x^5} dx$$

$$\text{e } \int_3^5 2\sqrt{t}(\sqrt{t} + 3(\sqrt{t})^3) dt$$

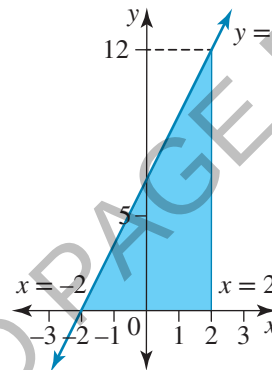
$$\text{f } \int_{-2}^{-1} \left(2 - \frac{4}{u^3} \right) du$$

10 a Determine the value of n so that $\int_4^n 3 dx = 9$.

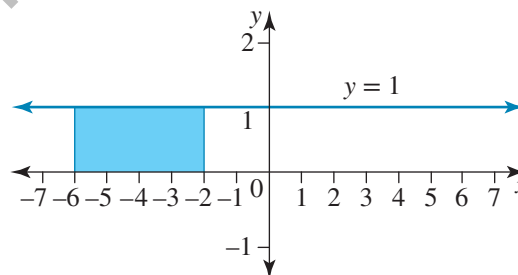
b Determine the value of p so that $\int_0^p \sqrt{x} dx = 18$.

11 Express the areas in each of the following diagrams in terms of a definite integral and calculate the area using both integration and a known formula for the area of the geometric shape.

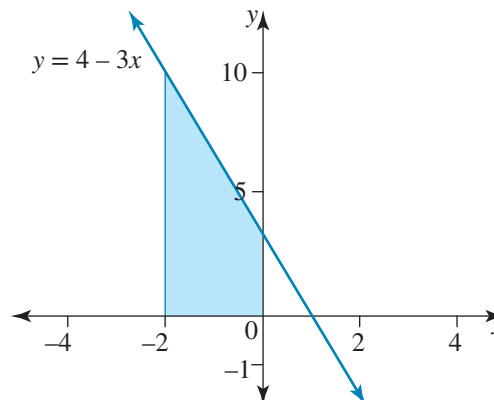
a The triangular area bounded by the line $y = 3x + 6$, the x -axis and $x = 2$ as illustrated in the diagram



b The rectangular area bounded by the line $y = 1$, the x -axis and $x = -6$, $x = -2$ as illustrated in the diagram



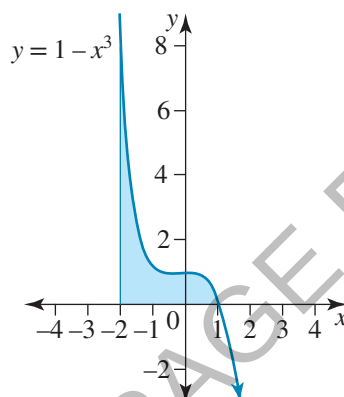
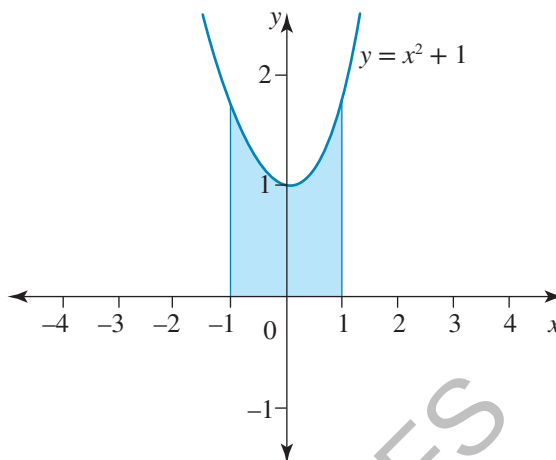
c The trapezoidal area bounded by the line $y = 4 - 3x$, the coordinate axes and $x = -2$ as illustrated in the diagram



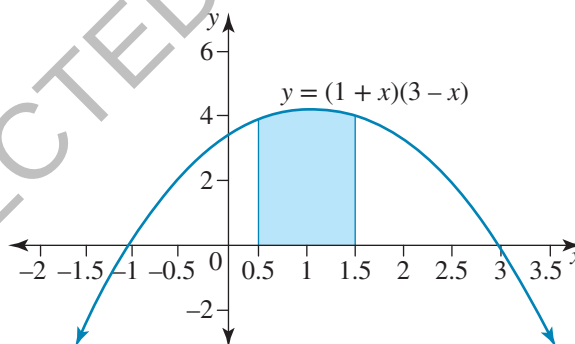
12 For each of the following areas:

- i write a definite integral, the value of which gives the area measure
- ii calculate the area.

- a** The area bounded by the curve $y = x^2 + 1$, the x -axis and $x = -1$, $x = 1$ as illustrated in the diagram
- b** The area bounded by the curve $y = 1 - x^3$, the x -axis and $x = -2$ as illustrated in the diagram



- c** The area bounded by the curve $y = (1 + x)(3 - x)$, the x -axis and $x = 0.5$, $x = 1.5$ as illustrated in the diagram



13 Sketch a graph to show each of the areas represented by the given definite integrals and then calculate the area.

a $\int_0^3 4x^2 dx$

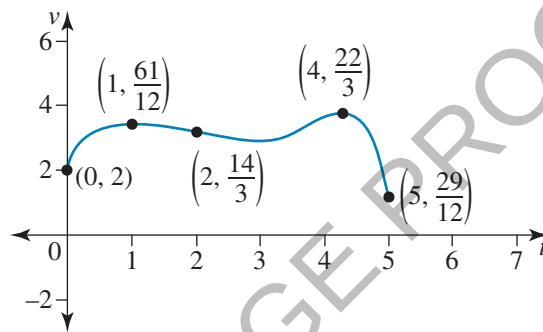
b $\int_{-1}^1 (1 - x^2) dx$

c $\int_{-2}^4 dx$

14 Consider the function defined by $f(x) = 4 - \sqrt{x}$.

- a** Sketch the graph of $y = f(x)$, showing the intercepts with the coordinate axes.
- b** On the diagram, shade the area which is bounded by the curve and the coordinate axes.
- c** Express the shaded area in terms of a definite integral.
- d** Calculate the area.

- 15** A particle starts from rest at the origin and moves in a straight line so that after t seconds its velocity is given by $v = 10t - 5t^2$, $t \geq 0$.
- Calculate an expression for the displacement at time t .
 - When the particle is next at rest, how far will it have travelled from its starting point?
 - Draw the velocity–time graph for $t \in [0, 2]$.
 - Use a definite integral to calculate the area enclosed by the velocity–time graph and the horizontal axis for $t \in [0, 2]$.
 - What does the area in part **d** measure?
- 16** An athlete training to compete in a marathon race runs along a straight road with velocity v km/h. The velocity–time graph over a 5-hour period for this athlete is shown below.



- What were the athlete's highest and lowest speeds over the 5-hour period?
- At what times did the athlete's acceleration become zero?
- Given the equation of the graph shown is $v = 2 + 8t - 7t^2 + \frac{7t^3}{3} - \frac{t^4}{4}$, use a definite integral to calculate the distance the athlete ran during this 5-hour training period.

MASTER

- 17** Write down the values of:

a $\int_{-3}^4 (2 - 3x + x^2) dx$

b $\int_4^{-3} (2 - 3x + x^2) dx$, and

- c** explain the relationship between parts **a** and **b**.

- 18 a** Use the calculator to obtain $\int_{-2}^3 (x + 2)(x - 3) dx$.

- b** Draw the graph of $y = (x + 2)(x - 3)$.

- c** What is the area enclosed between the graph of $y = (x + 2)(x - 3)$ and the x -axis?

- d** Why do the answers to part **a** and part **c** differ? Write down a definite integral which does give the area.



The Maths Quest Review is available in a customisable format for you to demonstrate your knowledge of this topic.

The Review contains:

- **Short-answer** questions — providing you with the opportunity to demonstrate the skills you have developed to efficiently answer questions without the use of CAS technology
- **Multiple-choice** questions — providing you with the opportunity to practise answering questions using CAS technology

- **Extended-response** questions — providing you with the opportunity to practise exam-style questions.

A summary of the key points covered in this topic is also available as a digital document.

REVIEW QUESTIONS

Download the Review questions document from the links found in the Resources section of your eBookPLUS.

Activities

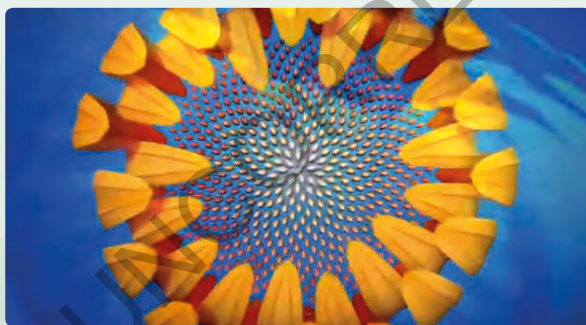
To access eBookPLUS activities, log on to



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Interactivities

A comprehensive set of relevant interactivities to bring difficult mathematical concepts to life can be found in the Resources section of your eBookPLUS.



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Units 1 & 2

Anti-differentiation and introduction to integral calculus



Sit topic test



14 Answers

EXERCISE 14.2

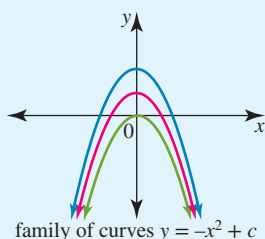
- 1 a $y = 2x^6$
 b $\frac{4}{3}x^3 + x^2 - 5x$
 c $f(x) = x^3 + x^2 - 16x + c$
- 2 $y = \frac{2x^3}{3} - \frac{3x^2}{2} + c$
- 3 a $-\frac{7x^5}{5} + x^3 - 3x^2 + c$
 b $\frac{5}{28}x^7 + \frac{3}{8}x^2 + c$
 c $F(x) = -\frac{2}{3x^3} + c$
 d $\frac{x^3}{3} + 2x - \frac{1}{x} + c$
- 4 a $F(x) = \sqrt{x} + c$
 b $\int (2t^2 + 4) dt = \frac{2t^3}{3} + 4t + c = 2 \int (t^2 + 2) dt$
- 5 a $y = \frac{1}{2}x^{10} + c$
 b $y = -3x + \frac{1}{2}x^8 + c$
 c $y = \frac{2}{3}x^3 - 6x^2 + 14x + c$
 d $y = -\frac{2}{3}x^3 + \frac{11}{2}x^2 + 40x + c$
- 6 a $f(x) = \frac{1}{12}x^6 + \frac{1}{3}x^7 + c$
 b $f(x) = \frac{4}{21}x^7 + 5x + c$
 c $f(x) = \frac{4}{3}x^3 - \frac{6}{7}x^7 + c$
 d $f(x) = 9x - 4x^3 + \frac{4x^5}{5} + c$
- 7 a $\frac{1}{15}x^9 + c$
 b $2x + c$
 c $40x^2 - \frac{5x^8}{2} + c$
 d $\frac{1}{100}[9x + 2x^3 - 0.5x^{11}] + c$
- 8 a $ax^2 + bx + c$
 b $0.0005x^{100} + c$
 c $2x^4 + 4x^3 + 3x^2 + x$
 d $7x - x^5 + \frac{4x^3}{3} + 4x^2 + c$
- 9 a $F(x) = \frac{1}{4}(x^3 - 2x) + c$
 b $F(x) = \frac{1}{24}x^2 + \frac{2}{3}x + c$
 c $F(x) = \frac{x}{4} + \frac{x^{15}}{12} + c$
 d $F(x) = \frac{4x^9}{9} - \frac{16x}{3} + c$
- 10 a $y = \frac{2}{5}x^{\frac{5}{2}} + c$
 b $y = -2x^{-\frac{1}{2}} + c$
- 11 a $f(x) = \frac{-5}{x} + c$
 b $f(x) = \frac{3x^4}{2} - 3x^{-2} + c$
- 12 a $\frac{x^4}{2} + \frac{7x^2}{2} + \frac{5}{x} + c$
 b $2x^2 + \frac{8}{3}x^{\frac{3}{2}} + x + c$
 c $\frac{5}{16}x^{\frac{16}{5}} - 10x^{\frac{13}{10}} + 10x^{\frac{6}{5}} + c$
 d $-\frac{1}{2x} - \frac{1}{2x^2} + \frac{4}{3x^3} + c$
- 13 a $F(x) = \frac{4a^2x^3}{3} + b^3x + c$
 b $F(x) = \frac{\sqrt{3}x^3}{3} + \sqrt{3}a^2x + c$
- 14 a $\frac{1}{4}x^4 + \frac{4}{3}x^3 - \frac{1}{2}x^2 - 4x + c$
 b $\frac{-25}{x} - 2x + \frac{x^3}{75} + c$
 c $\frac{1}{p-q+1}x^{p-q+1}, p \neq q-1$
 d $3x^{\frac{1}{3}} - \frac{3}{5}x^{\frac{5}{3}}$
 e $4x + 7$
- 15 $\frac{x^4}{4} + c$
- 16 a $\frac{2x^{\frac{5}{2}}}{5}$
 b $\frac{x^9}{9} + \frac{2x^5}{5} + x$
 c $50t^2 + \frac{500}{t}$
 d $\frac{2y^{10} + 3}{12y^4}$
 e $\frac{(4u+5)^{\frac{3}{2}}}{6}$
 f Answers will vary.

EXERCISE 14.3

- 1 a $y = -3x^2 - 6x + 7$
 b $f(-1) = 2$
- 2 $y = \frac{2}{3}$

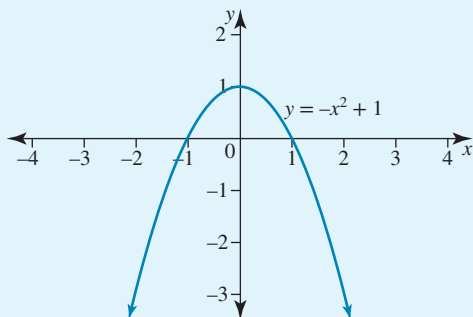
3 a Maximum turning point at $x = 0$

b



c $f(x) = -2x$

d $F(x) = -x^2 + 1$;



4 $y = \frac{1}{4}x^2 + 2x + 6$; minimum turning point at $(-4, 2)$

5 $f(x) = -x^3 + 4x + 5$

6 a $y = \frac{x^2}{5} - 3x + 10$

b $(5, 0), (10, 0)$

7 a $k = -\frac{3}{2}$

b $y = -\frac{3x^2}{4} + 8$

8 $F(x) = \frac{-2}{3x^3} + \frac{9}{20x^2} - \frac{1}{10x} - \frac{41}{60}$

9 a $y = -9$

b $z = 128$

c $A = 72$

d $y = \frac{1}{4}$

10 a $a = 4$

b $y = 4x - \frac{3x^{\frac{5}{3}}}{5} + \frac{96}{5}$

c $5y - 15x = 98$

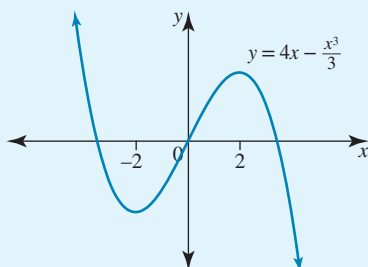
11 a $a = 6$

b $f(x) = \frac{6}{x} + 6$

c 81°

12 a Minimum turning point at $x = -2$; maximum turning point at $x = 2$

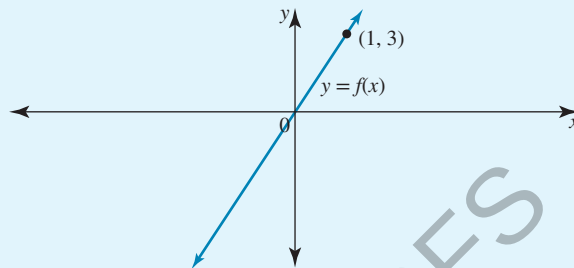
b



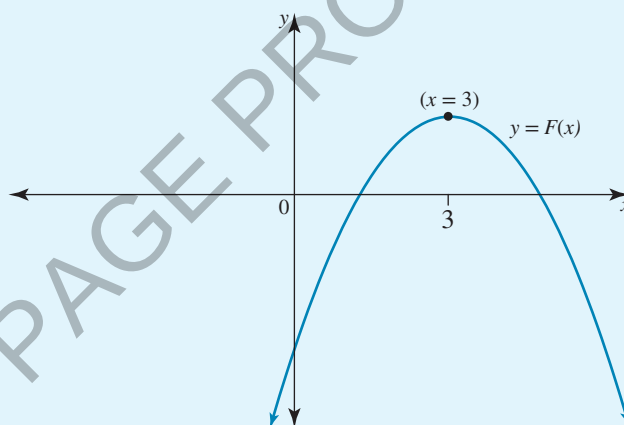
c $\frac{dy}{dx} = 4 - x^2$; $y = 4x - \frac{x^3}{3} + c$

d $y = -\frac{x^3}{3} + 4x - 3$; slope of -5

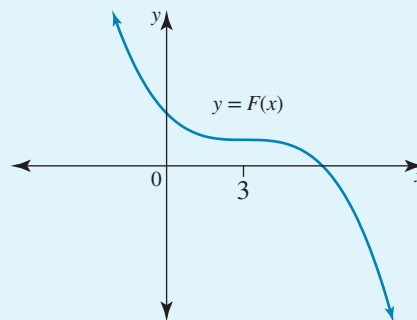
13 a Any line with gradient of 3



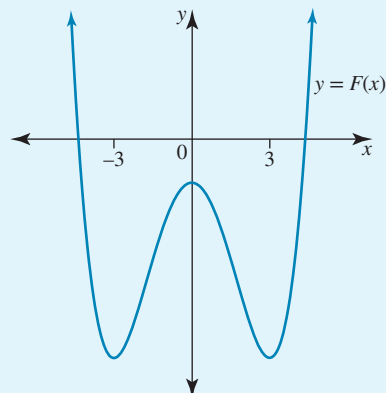
b Any parabola with a maximum turning point when $x = 3$



c Any inverted cubic with a stationary point of inflection when $x = 3$



d Any quartic with minimum turning points when $x = \pm 3$ and a maximum turning point when $x = 0$



14 a $-\frac{9x^4}{4} + 18x + c$

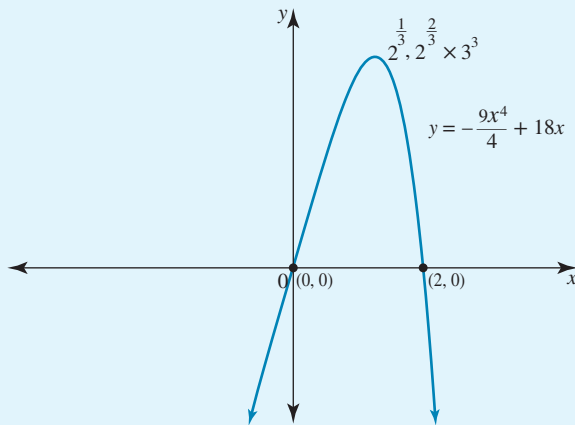
b i $y = -\frac{9x^4}{4} + 18x$

ii (2, 0); steeper at (2, 0)

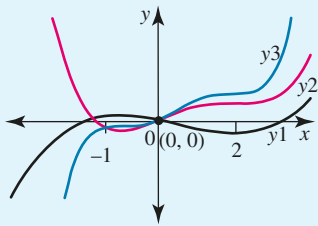
iii Maximum turning point when $x = \sqrt[3]{2}$

iv $p = -\frac{2}{3}, q = 3$

v



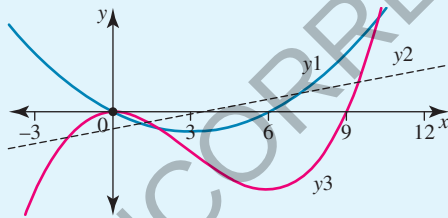
15 a,b,c Use your CAS technology to sketch the graphs.



d For factors of multiplicity 1, the anti-derivative graph has turning points at the zeros of each factor.

For factors of multiplicity 2, the anti-derivative graph has stationary points of inflection at the zeros of each factor.

16 Use your CAS technology to sketch the graphs.



The x-intercept of $y = f'(x)$ is connected to the turning point of $y = f(x)$.

The turning points of $y = \int f(x) dx$ are connected to the x-intercepts of $y = f(x)$.

EXERCISE 14.4

1 a $2\sqrt{2}$ seconds

b 14 m/s^2

2 a $-\frac{2}{t^3}, x = -\frac{1}{t} + 2t - 2$

3 Velocity 9 m/s; position 9 metres to the right of the origin

4 122.5 metres

5 a $V = -0.25t + 4.5\pi$

b 56.5 seconds

6 64.4 cm

7 a $x = \frac{8t^3}{3} - 10t^2 - 12t + 54$

b $19\frac{1}{3}$ metres

c At the origin

8 13 metres to the right of the origin

9 a $a = -6$

b 1 second; 12 metres to the right of origin

c 15 metres

d 5 m/s

e -3 m/s

10 a $24 \text{ m/s}; -18 \text{ m/s}^2$

b $x(t) = t^3 - 9t^2 + 24t$

c 20

d 28 metres

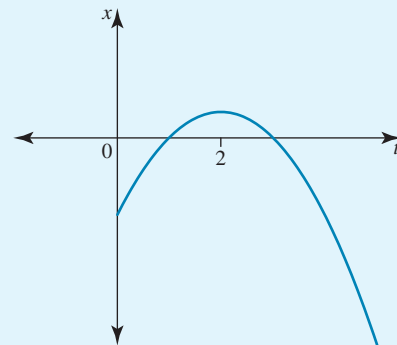
11 a $v = 3t^2 + 8t - 8$

b $x = t^3 + 4t^2 - 8t + 5$

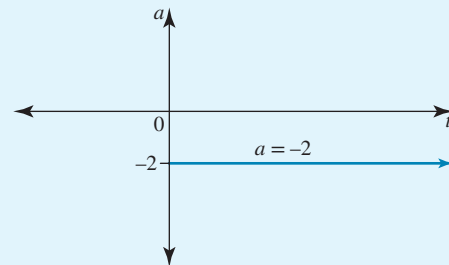
12 a 2 seconds; 20 metres to right of origin

b 4 seconds

13 a Part of a parabola with a maximum turning point at $t = 2$

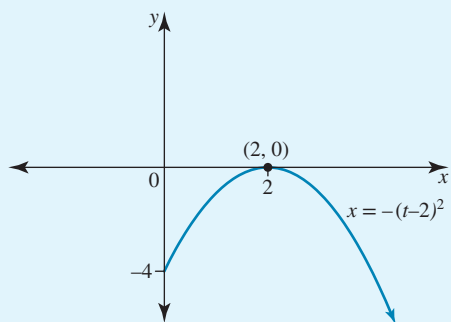


b Part of a horizontal line below the t -axis through (0, -2)



c $x = -(t-2)^2, t \geq 0$

d Part of a parabola with maximum turning point at $(2, 0)$



14 a $A = -9t^2 + 90$

b 4 days

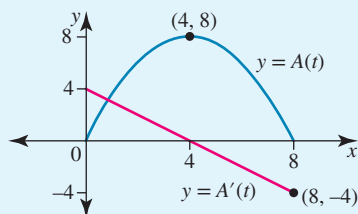
c $0 \leq t \leq \sqrt{10}$

15 a 4 days

b $A(t) = 4t - \frac{t^2}{2}$; domain $[0, 8]$

c 8 m^2

d



16 a $k = 150$

b $m = 100t^{\frac{3}{2}} + 20$

c 16 days

17 a $a = 8(2t + 1)^3$

b $x = \frac{(2t + 1)^5}{10} + 4.1$

c 0.56 seconds; 20.27 m/s

18 $x = \frac{t}{2} + \frac{1}{2(t+1)} - \frac{1}{2}$

EXERCISE 14.5

1 18

2 a $-\frac{32}{3}$

b 0

3 a 6 square units

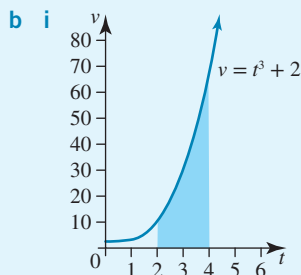
b $\int_0^4 0.75x \, dx$

c 6 square units

4 a $\int_1^3 (16 - x^2) \, dx$

b $23\frac{1}{3}$ square units

5 a $v = 3 \text{ m/s}; a = 3 \text{ m/s}^2$



ii $\int_2^4 (t^3 + 2) \, dt = 64$ so distance is 64 metres.

6 a Velocity is a positive definite quadratic function in t .

b i 14 metres

ii 14 metres

7 a 64

b 6

c 70

d -351

e $\frac{8}{3}$

f 0

8 Proofs required

9 a $\frac{1}{2}$

b 18

c 62

d $\frac{55}{16}$

e 212

f $\frac{7}{2}$

10 a $n = 7$

b $p = 9$

11 a $\int_{-2}^2 (3x + 6) \, dx$; 24 square units

b $\int_{-6}^{-2} dx$; 4 square units

c $\int_{-2}^0 (4 - 3x) \, dx$; 14 square units

12 a i $\int_{-1}^1 (x^2 + 1) \, dx$

ii $2\frac{2}{3}$ square units

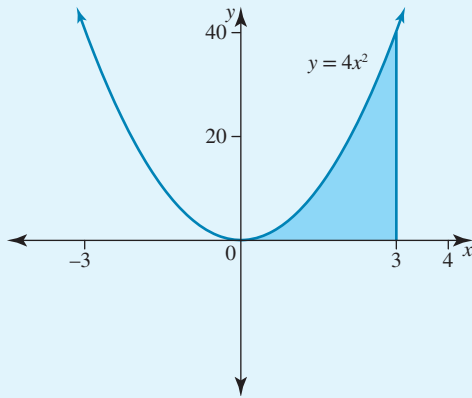
b i $\int_{-2}^1 (1 - x^3) \, dx$

ii $6\frac{3}{4}$ square units

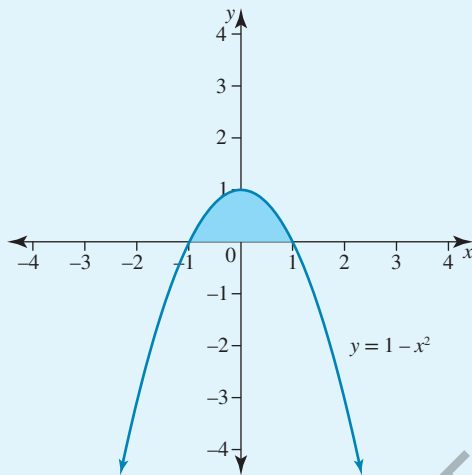
c i $\int_{0.5}^{1.5} (1 + x)(3 - x) \, dx$

ii $3\frac{11}{12}$ square units

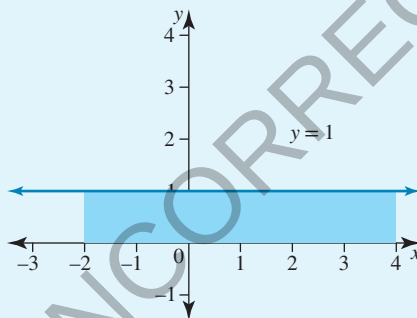
- 13 a** Area enclosed by $y = 4x^2$, x -axis and $x = 0$, $x = 3$ is 36 square units.



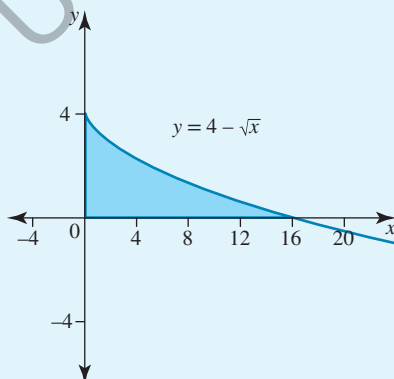
- b** Area enclosed by $y = 1 - x^2$ and x -axis is $\frac{4}{3}$ square units.



- c** Rectangular area enclosed by the line $y = 1$, x -axis and $x = -2$, $x = 4$ is 6 square units.



- 14 a b** Intercepts $(0, 4)$, $(16, 0)$;

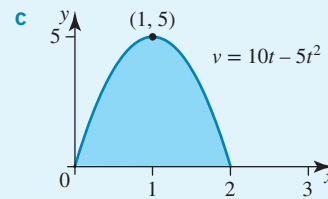


- c** Area measure equals $\int_0^{16} (4 - \sqrt{x}) dx$.

- d** $\frac{64}{3}$ square units

- 15 a** $x = 5t - \frac{5t^3}{3}$

- b** $6\frac{2}{3}$ metres



- d** $6\frac{2}{3}$

- e** Distance travelled in first 2 seconds.

- 16 a** $7\frac{1}{3}$ km/h; 2 km/h

- b** 1 hour, 2 hours and 4 hours

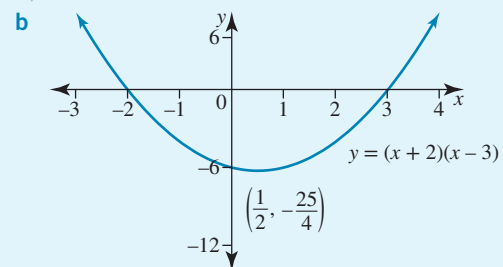
- c** $26\frac{2}{3}$ km

- 17 a** $\frac{203}{6}$

- b** $\frac{203}{6}$

- c** Interchanging the terminals changes the sign of the integral.

- 18 a** $-\frac{125}{6}$



- c** $\frac{125}{6}$ square units

- d** Area lies below x -axis so signed area is negative.

Possible integrals are $-\int_{-2}^3 (x+2)(x-3) dx$ or

$$\int_3^{-2} (x+2)(x-3) dx.$$